What is Model Context Protocol?
Ek standard hai jo AI models (jaise LLMs) ko external tools, APIs, databases aur systems ke saath connect karta hai — *smart tareeke se*.
What is a REST API?
It allows applications (websites, mobile apps, servers, AI tools, etc.) to **send requests and receive data** from another system.
FastMCP Server?
FastMCP Server aik special server hota hai jo AI models (jaise OpenAI ke models) ko external tools, APIs, databases ke sath connect karta hai through (MCP).
FastMCP Client?
FastMCP Client wo "waiter" hai jo AI (customer) ki request ko uthata hai aur **FastMCP Server (kitchen)** tak pohanchata hai.
MCP Protocol Methods?
MCP ke *methods* woh specific actions hain jo AI system use karta hai communication ke liye — jaise client aur server ke darmiyan "commands".
MCP Tools?
MCP Tools wo external functions ya services hoti hain jo AI ko *real duniya ke kaam karne ki power deti hain*.
MCP Resources?
MCP Resources = wo data, files, APIs, documents ya tools ka content jo AI use karta hai kaam karne ke liye
MCP Prompts?
MCP Prompts woh *instructions / recipes* hoti hain jo AI ko batati hain.
JSON-RPC — The Language of MCP?
JSON-RPC ek lightweight communication protocol hai jo client aur server ke darmiyan baat karne ka tareeqa define karta hai — using JSON format.
MCP Sampling ka matlab hai ke jab AI (jaise LLM) response generate karta hai, to woh ek hi fixed jawab nahi deta — balkay multiple possible answers me se **smart selection (sampling)** karta hai.
MCP Transports ka matlab hai woh *raaste (channels)* jahan se **Client aur Server ke darmiyan data travel karta hai** in **Model Context Protocol (MCP)**.
SSE (Server-Sent Events)
Server (chef) khud se continuously updates bhejta rehta hai client ko (customer ko) — bina baar baar request ke.

What is Model Context Protocol?

A protocol that allows AI models to securely connect to external tools, data sources, and systems.

Model Context Protocol (MCP) ek standard hai jo AI models (jaise LLMs) ko external tools, APIs, databases aur systems ke saath connect karta hai — *smart tareeke se*
MCP = AI ka "waiter" jo kitchen (tools/APIs) aur customer (AI model) ke beech communication manage karta hai.

Why we use MCP?

Without MCP:

- AI sirf apni training tak limited hota hai
- Real-time data access nahi hota
- Tools use karna mushkil hota

With MCP:

- AI real-time data le sakta hai
- APIs call kar sakta hai
- Tools use kar sakta hai
- Smart decisions le sakta hai

How we use MCP?

MCP ka flow kuch is tarah hota hai:
Steps:

- User question deta hai
- AI samajhta hai ke kya chahiye
- MCP Client request bhejta hai
- MCP Server decide karta hai kaunsa tool use hoga
- Tool/API se data aata hai
- AI response bana ke user ko deta hai

Why MCP is Important in AI?

- Har tool ke liye alag integration nahi karna padta
- AI ko relevant context milta hai
- AI khud tools use karta hai (agent behavior)
- Chatbots, AI employees, Automation systems, RAG (Retrieval Augmented Generation)

Biryani Style Example

Socho tum ek **biryani restaurant** chala rahe ho

Components Mapping:

- **Customer** → User
- **Waiter** → MCP
- **Chef** → AI Model
- **Kitchen tools** → APIs / Database

Customer kehta hai:

"Mujhe spicy chicken biryani chahiye"

Without MCP

Chef confuse ho jata hai:

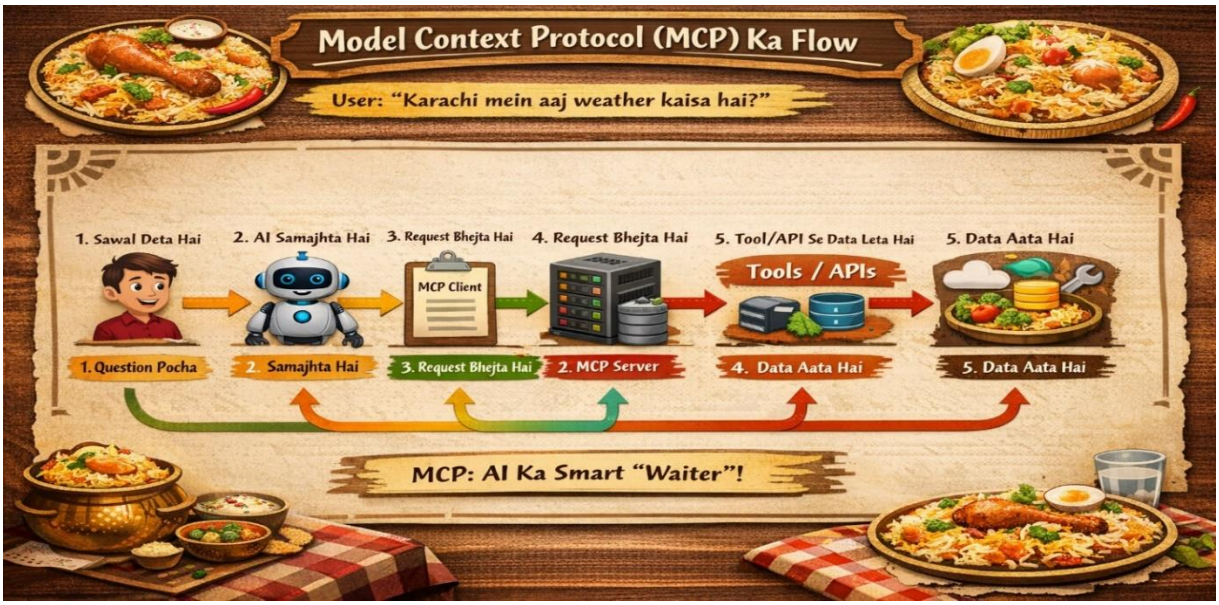
- Spice kitna?
- Rice available hai ya nahi?
- Chicken fresh hai ya nahi?

Result: ya toh galat order ya delay

With MCP

- Waiter (MCP) order leta hai
- Kitchen se check karta hai:
  - Chicken available? ✔
  - Rice ready? ✔
  - Spice level? ✔
- Chef ko proper context deta hai
- Chef perfect biryani banata hai

Result: **perfect response**



What is a REST API?

A **REST API** is a way for **two computer systems to communicate over the internet using standard HTTP methods**.

It allows applications (websites, mobile apps, servers, AI tools, etc.) to **send requests and receive data** from another system.

Think of a REST API as a **waiter in a restaurant** 🍽️:

- **Client (You)** → The person ordering food
- **API (Waiter)** → Takes the request to the kitchen
- **Server (Kitchen)** → Prepares the food
- **Response (Food served)** → The result returned to you

1️⃣ Endpoint

An **endpoint** is a URL where the API receives requests.

2️⃣ HTTP Methods (Actions)

REST APIs use standard **HTTP methods**:

Method	Purpose	Example
GET	Retrieve data	Get list of products
POST	Create new data	Add a new product
PUT	Update data	Update product details
DELETE	Remove data	Delete a product

3️⃣ **Request**

The **client sends a request** to the server

4️⃣ **Response**

The server sends data back, usually in **JSON format**.

```
{
"id": 10,
"name": "Laptop",
"price": 750,
"stock": 12
}
```

What FastMCP Server?

FastMCP Server is a lightweight Python framework used to quickly build an MCP server that allows AI models to connect with tools, resources, and external systems using the Model Context Protocol (MCP).

FastMCP is basically the server layer that exposes tools to AI.
Install FastMCP
pip install fastmcp
fastmcp run main.py --transport sse --port 8000

Test with MCP Inspector
npx @modelcontextprotocol/inspector

Run the Client
python client.py

FastMCP Server aik special server hota hai jo AI models (jaise OpenAI ke models) ko external tools, APIs, databases ke sath connect karta hai through Model Context Protocol (MCP).

FastMCP Server = "AI ka waiter" jo AI ko tools serve karta hai.

Biryani Style Explanation

AI = Guest
FastMCP Server = Waiter
Tools/APIs = Biryani ke dishes

- Guest (AI) bolta hai:
  "Mujhe user ka data chahiye"
- Waiter (FastMCP Server) kya karta hai?
  Kitchen (API/DB) se data laata hai
  Plate mein serve karta hai
- AI ko khud kitchen mein nahi jana parta!

FastMCP Server kyun use karte hain?

- AI directly APIs call nahi karta — FastMCP Server karta hai.
- API keys server mein safe rehti hain (AI ko direct access nahi milta).
- MCP protocol use hota hai → clean, standard communication.
- AI ko powerful banane ke liye database read, APIs call, automation.

FastMCP Server kaise use karte hain?

Step-by-step flow:

- AI Agent request bhejta hai
- MCP Client request forward karta hai
- FastMCP Server receive karta hai
- Server tool/API call karta hai
- Response AI ko wapas deta hai

Karachi Traffic Challan Check Example

User bolta hai:

"Mera challan check karo"

AI bolta hai: "Number plate KA1234 ka data lao"

FastMCP Server: "Traffic API call karta hai."

API response: "Rs 2000 challan pending"

Server: AI ko deta hai:

AI: User ko bolta hai "Aapka challan Rs 2000 hai"

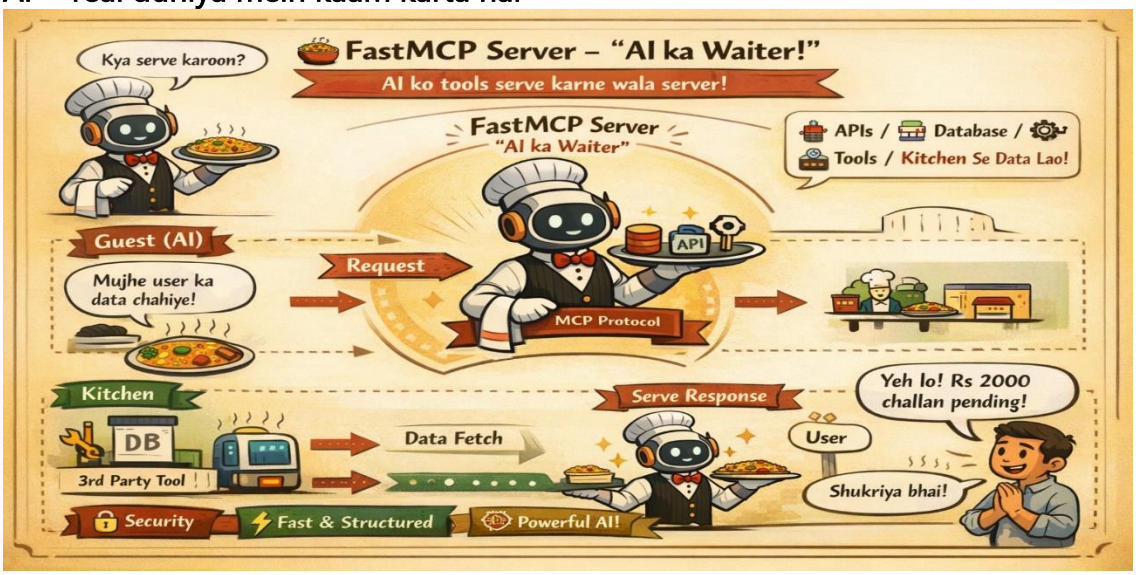
AI mein FastMCP Server important kyun hai?

Without FastMCP:

AI = sirf baatein karta hai (no real action)

With FastMCP:

AI = real duniya mein kaam karta hai



What is FastMCP Client?

Socho tum AI ka system bana rahe ho jahan AI agent ko bahar ke tools (APIs, DB, functions) se baat karni hai.
FastMCP Client wo “waiter” hai jo AI (customer) ki request ko uthata hai aur FastMCP Server (kitchen) tak pohanchata hai.

Simple line:
FastMCP Client = AI aur tools ke darmiyan communication bridge

Biryani Example (Zabardast Samajh)

- Kitchen = FastMCP Server
- Customer = AI Agent
- Waiter = FastMCP Client

Customer (AI) bolta hai:
“Mujhe Chicken Biryani chahiye”
Waiter (Client):
Order le kar kitchen (Server) ko deta hai
Kitchen:
Biryani bana kar wapas deta hai
Waiter:
Customer ko serve karta hai
Yani Client ka kaam hai:

- Request lena
- Server ko bhejna
- Response wapas lana

FastMCP Client kyun use karte hain?
AI directly har API se baat nahi karta → Client beech mein handle karta hai

FastMCP optimized hota hai → low latency

- Sab tools ek hi language mein baat karte hain (MCP protocol)
- Client change karo → server ya tools change karne ki zarurat nahi

FastMCP Client kaise use karte hain?

- AI Agent request banata hai
- FastMCP Client us request ko MCP format mein convert karta hai
- Server ko send karta hai
- Server tool run karta hai
- Response client ko milta hai
- Client AI ko return karta hai

coding

```
from fastmcp import Client
client = Client("http://localhost:8000")
response = client.call_tool(
name="get_weather",
arguments={"city": "Karachi"})
print(response)
```

AI mein FastMCP Client important kyun hai?

- AI sirf text nahi → real world actions kar sakta hai
- New tools easily add ho jate hain
- Multi-tools integration
- Modern AI Agents (AutoGPT style) isi architecture pe depend karte hain

MCP Protocol Methods

MCP (Model Context Protocol) ke *methods* woh specific actions hain jo AI system use karta hai communication ke liye — jaise client aur server ke darmiyan "commands".
Methods = woh tareeqay jin se AI baat karta hai, data mangta hai, aur kaam karta hai"

Biryani Style Samajh lo
Socho tum ek **biryani restaurant** chala rahe ho:

- **Server (Kitchen)** = Tools / APIs
- **Client (Waiter)** = AI Agent
- **User (Customer)** = Tum

Ab **MCP Methods = Waiter ke actions**
Important MCP Protocol Methods

1. initialize
System start karne ke liye
"Bhai waiter kitchen ko bol raha: boss ready ho jao, customer aa gaya hai!"

2. tools/list
Available tools ki list lena
"Waiter: aaj menu mein kya kya hai? Chicken biryani, beef biryani ya pulao?"

3. tools/call
"Customer ne bola: ek spicy chicken biryani lao 🔥
Waiter kitchen ko order deta hai → yehi tools/call hai".

4. resources/list
Available data/resources dekhna
"Kitchen check: chawal hai? masalay hain? chicken fresh hai?

5. resources/read
Specific data read karna
"Recipe book kholo aur dekho chicken biryani ka exact formula kya hai"

6. completion / response
Final answer dena
Waiter garam garam biryani table pe serve karta hai

MCP Protocol Methods kyun use karte hain?

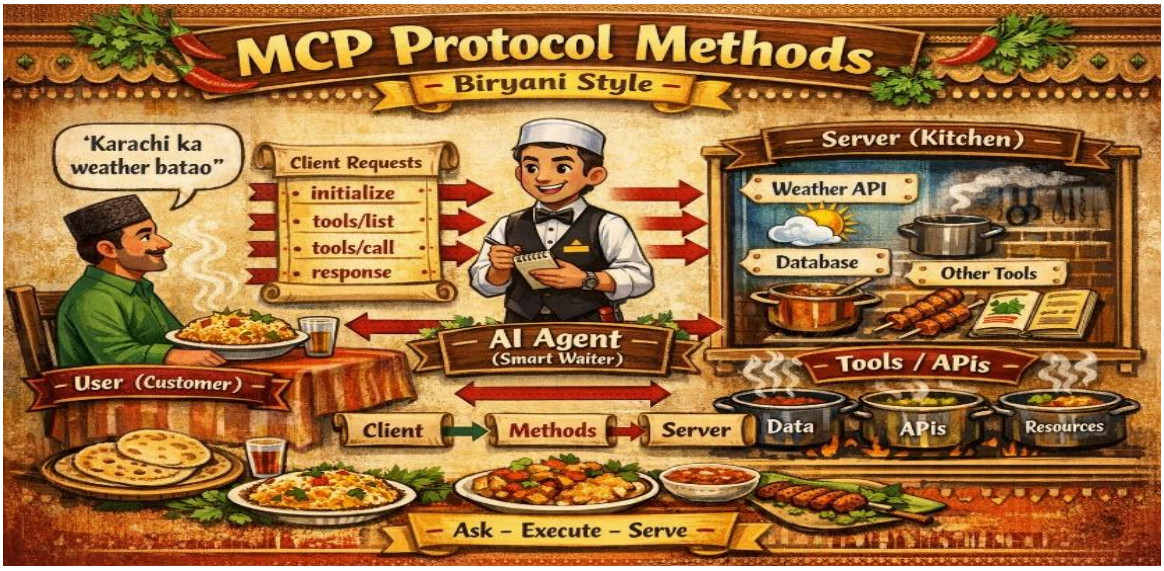
- AI ko clear instructions milti hain
- AI external tools use kar sakta hai
- Multiple tools handle kar sakta hai

MCP Methods ka use kaise hota hai?

- initialize → system start
- tools/list → tools check
- tools/call → kaam karwana
- response → output dena

AI mein MCP Methods kyun important hain?

- AI ko real-world se connect karta hai
- Automation easy hoti hai
- Modular system banata hai



MCP Tools (Model-Controlled Primitive)

Tools are executable functions exposed by an MCP server that allow AI models to perform actions such as creating, updating, or deleting data in external systems.
They follow a model-controlled paradigm — the AI model autonomously discovers available tools via tools/list and decides when to invoke them via tools/call.

MCP (Model Context Protocol) Tools wo external functions ya services hoti hain jo AI ko real duniya ke kaam karne ki power deti hain.

AI khud sirf soch sakta hai... **Tools usay kaam karne ke liye haath dete hain**

Biryani Style Explanation
Socho tum biryani bana rahe ho

- AI = Chef
- MCP = Recipe system
- MCP Tools = Ingredients + kitchen tools

Agar chef ke paas sirf recipe ho aur ingredients na ho... toh biryani nahi banegi
AI bina MCP Tools ke sirf baat kar sakta hai, kaam nahi

MCP Tools kyun use karte hain?

- Live data laata hai (weather, stock, DB)
- Email bhejna, file banana, API call karna
- Guess nahi karta, actual data use karta hai
- AI sirf chatbot nahi, *assistant* + *worker* ban jata hai

MCP Tools kaise use hotay hain?

- User bolta hai: "Karachi ka weather batao"
- AI samajhta hai: Mujhe weather tool use karna hai
- MCP Client request bhejta hai
- MCP Server tool ko call karta hai
- Tool result deta hai
- AI user ko jawab deta hai

MCP Tools AI mein important kyun hain?
Without MCP Tools:
AI = sirf talking machine
With MCP Tools:
AI = Smart Assistant + Worker + Decision Maker

MCP TOOLS
Karachi ka weather batao!
32°C Kal 10 AM Meeting Set!
Request Sent
Result Received
Weather API
Database Query
Web Search
Calendar Tool
AI AGENT
HAPPY USER
AI Calls Tools
Tools Send Result

MCP Resources (App-Controlled Primitive)

Resources are read-only data endpoints exposed by an MCP server, allowing AI models to access pre-authorized information such as documents, configurations, and reference data.

They follow an app-controlled paradigm — the host application decides which data to expose, and the model simply reads what is made available
via resources/list and resources/read.

Socho tum biryani bana rahay ho...

- Chawal, gosht, masalay, oil → yeh sab kya hain?
  ya Resources (ingredients) hai.

Isi tarah AI system mein:
MCP Resources = wo data, files, APIs, documents ya tools ka content jo AI use karta hai kaam karne ke liye

MCP Resources wo cheezain hain jahan se AI knowledge ya data leta hai.

MCP Resources kyun use hotay hain?

- Agar biryani mein sirf chawal ho...
- taste nahi ayega
  value nahi hogi

Isi tarah AI:

- bina data ke useless
  bina resources ke answer nahi de sakta
- Is liye MCP Resources use hotay hain:
- Data provide karne ke liye
- Context dene ke liye
- Real-world info lane ke liye
- AI ko smart banane ke liye

MCP Resources kaise use hotay hain?
AI ka flow kuch is tarah hota hai:
🤵 (AI Agent) → 👨🍳 (MCP Client) → 🍲 (MCP Server) → 🍘 (Resources)
Step-by-step biryani style:

1. User bolta: “Mujhe user ka data chahiye”
2. AI Agent request bhejta
3. MCP Client us request ko handle karta
4. MCP Server jaata hai Resource ke paas
5. Resource (DB/API/file) se data uthata
6. Wapas AI ko deta
7. AI final answer bana deta

MCP Resources AI mein itna important kyun hain?
AI + Resources = Powerful System
AI – Resources = Sirf theory

MCP Prompts (User-Controlled Primitive)

Prompts are pre-crafted instruction templates created by domain experts that encode specialized knowledge into reusable, discoverable workflows.
They follow a user-controlled paradigm — the user explicitly selects which prompt to apply (e.g., via slash commands like /contract_review), unlike tools where the model decides autonomously.
MCP Prompts (Model Context Protocol Prompts) woh *instructions / recipes* hoti hain jo AI ko batati hain:
"Tum kya karo?"
"Kaise karo?"
"Kis data ke sath karo?"

Why we use MCP Prompts?

- Socho agar biryani bina recipe ke banaoge…
- kabhi namak zyada
- kabhi chawal kaccha
- kabhi masala missing

Isi tarah AI bhi bina prompt ke:

- galat jawab
  incomplete answer
  random output
- Is liye MCP Prompts use karte hain taake AI:
  ✓ clear samjhay
  ✓ correct output de
  ✓ structured kaam kare

How we use MCP Prompts?

Step 1: Ingredients define karo (Context)

- Data kya hai?
- Problem kya hai?

Step 2: Instructions do (Prompt)

- AI ko clear task do

Step 3: Cooking process (Execution)
AI process Karega
Step 4: Output serve
Final answer mil jata hai

Why MCP Prompts important in AI?

- AI ek powerful chef hai 👨‍🍳
- Lekin bina instructions ke confuse ho jata hai.

MCP Prompts ki importance:

- AI ko direction dete hain
- Consistency maintain karte hain
- Complex systems (Agents, Tools, APIs) ko control karte hain
- Automation possible banate hain

Without MCP Prompt:
"Biryani banao"
AI sochega:

- Karachi style?
- Hyderabadi?
- Chicken ya beef?

Result: Confusion

With MCP Prompt (Perfect Recipe):

- "Chicken biryani banao
- 2 logon ke liye
- medium spicy
- step by step instructions do aur ingredients bhi batao"

JSON-RPC — The Language of MCP?

What is JSON-RPC?
Think of your MCP Client as a person walking into a shop (MCP Server) to ask for something. The way they ask and receive — that specific language — is JSON-RPC.

1 Request
The client asks the server to do something

2 Response
The server replies back.

3 Error
If something goes wrong:

4 Notification (Bonus)
A notification has **no id**. The client just informs the server — no reply needed.

5 Simple Analogy

JSON-RPC	Real Life
Request	Placing a pizza order
id	Your order/token number
method	Which pizza you want
result	Pizza delivered
error	Pizza not available
notification	"I'm on my way" — no reply expected

JSON-RPC ek lightweight communication protocol hai jo **client aur server ke darmiyan baat karne ka tareeqa** define karta hai — using **JSON format**.

“JSON-RPC = Message format jisme AI Agent MCP server se baat karta hai”

Biryani Style Samajh Lo

- Rice = JSON (data format)
- Chicken = RPC (Remote Procedure Call)
- Masala = Method + Params
- Dum = Response

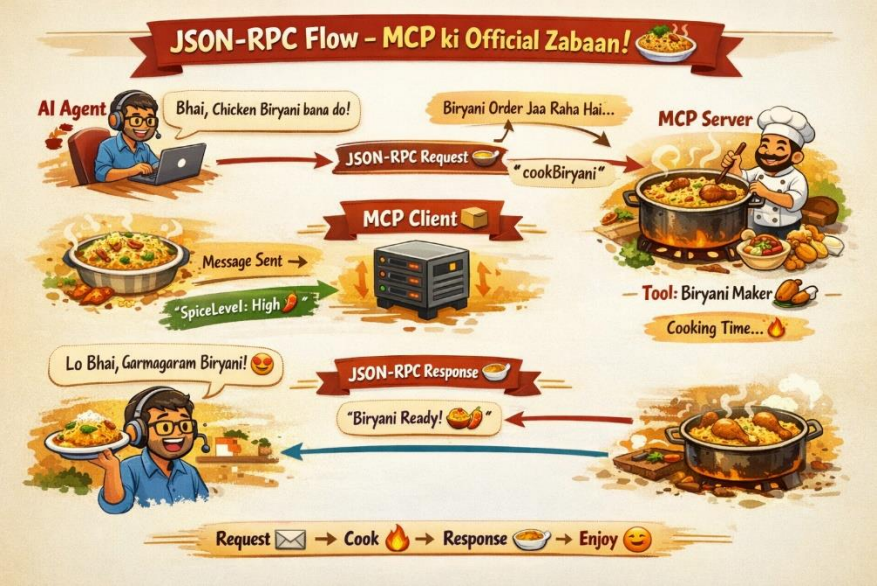
Result:
Perfect Biryani = JSON + RPC = JSON-RPC

How we use JSON-RPC?
{"jsonrpc": "2.0","method": "cookBiryani","params": {"type": "chicken","spiceLevel": "high"}, "id": 101}

Server ka response:
{ "jsonrpc": "2.0", "result": " Biryani Ready", "id": 1}

MCP me iska role

- Agent bolta hai: "tool call karo"
- JSON-RPC format me message jata hai
- Har MCP tool same language samajhta hai
- AI automatically tools use karta hai
- Minimal data → fast response



MCP Sampling

In the normal flow:
Client → Server → Result

In sampling: **Server → Client → "Ask the LLM" → Server**

The server wants to use an LLM itself — but it can't do it directly, so it asks the client!

MCP Sampling ka matlab hai ke jab AI (jaise LLM) response generate karta hai, to woh ek hi fixed jawab nahi deta — balkay multiple possible answers me se **smart selection (sampling)** karta hai.

AI ke paas ek *degchi full of options* hoti hai... aur woh decide karta hai kaunsa chawal (token/word) uthana hai

Socho tum biryani bana rahe ho:

- Ingredients = Context (MCP context)
- Masalay = Prompts
- Chef = AI Model
- Taste adjust karna = **Sampling**

Without Sampling:
Agar chef har dafa same recipe follow kare:
Taste boring ho jata hai (deterministic)
With Sampling:
Lekin agar thora variation kare:
Kabhi spicy, kabhi mild = **Sampling**
Why we use MCP Sampling?
Reason 1: Natural Responses
Agar sampling na ho:

- AI har dafa same answer dega

Sampling ke sath:

- AI human-like aur creative ho jata hai

How we use MCP Sampling?

User Prompt deta hai

"Mujhe ek funny Python song likho"

MCP Context ready hota hai

Tools, memory, data add hota hai

Model probabilities banata hai

Har next word ka chance calculate hota hai

Sampling apply hoti hai

Best words me se ek choose hota hai

Final output generate hota hai

Funny song ready!

Why MCP Sampling is Important in AI?

- AI robotic nahi lagta — natural lagta hai
- Har response fresh hota hai
- Coding → low sampling (accurate)
- Songs/Stories → high sampling (creative)
- AI agents jab decisions lete hain:
- Multiple options explore karte hain
- Best action choose karte hain

MCP Transports

A **transport** is the method used to send messages between the Client and Server. Think of it as the road between two houses — the houses are the same, but the road can be different.

MCP Transports ka matlab hai woh *raaste (channels)* jahan se **Client aur Server ke darmiyan data travel karta hai** in **Model Context Protocol (MCP)**.

Transport = Data bhejne aur receive karne ka tareeqa

Common MCP Transports:

- HTTP (REST style)
- WebSocket (real-time)
- STDIO (local communication)
- SSE (Server-Sent Events)

Socho tum biryani bana rahe ho

- **Chef (AI Agent)** → order deta hai
- **Waiter (Client)** → message le kar jata hai
- **Kitchen (Server)** → biryani banata hai
- **Transport (raasta)** → waiter ka chalne ka tareeqa

Agar waiter:

- bike pe jaye = HTTP 🚵
- walk kare = STDIO 🚶
- live call kare = WebSocket 📞

Toh yeh sab **MCP Transports** hain

Why We Use MCP Transports?

Hum MCP Transports use karte hain kyunki:

1. Communication enable hota hai
    - Client aur Server baat kar sakte hain
2. Different environments support
    - Local machine → STDIO
    - Internet → HTTP/WebSocket
3. Speed & efficiency control
    - Real-time → WebSocket
    - Simple request → HTTP

How We Use MCP Transports?

Step-by-step flow:

1. AI Agent request banata hai
2. Client us request ko transport ke through bhejta hai
3. Server request process karta hai
4. Response wapas same transport se aata hai

Example:

- Client → HTTP request bhejta hai
- Server → JSON-RPC response deta hai
- Result → AI ko mil jata hai

Types of MCP Transports

1. HTTP (Sabse common)

Use case: normal AI queries

2. WebSocket (Real-time king 👑)

Use case: chatbots, live AI agents

3. STDIO (Local ninja)

Use case: local tools integration

4. SSE (Server-Sent Events)

Use case: live logs, AI streaming responses

Why MCP Transports Important in AI?

1. AI ko real duniya se connect karta hai
2. Tools aur APIs integrate karta hai
3. Fast aur scalable communication deta hai
4. Different platforms support karta hai

Without transport = AI agent akela

With transport = AI agent powerful

SSE (Server-Sent Events)

SSE means the server **runs independently on a port** (like a website) and clients **connect to it over HTTP**. The server stays alive and keeps an open connection so it can push messages back to the client in real-time.

Socho tum ek **biryani hotel** chala rahe ho
Aur customer (browser/app) table par baitha hai.

Server (chef) khud se continuously updates bhejta rehta hai client ko (customer ko) — bina baar baar request ke.

Server-Sent Events ek technology hai jahan server continuously data push karta hai client ko ek open connection par.

2. Why we use SSE?
Socho agar har dafa customer waiter ko bula kar bole:
"Bhai biryani ready hui?"
Ye inefficient hai!
SSE solve karta hai:

- Bar bar request ki zaroorat nahi
- Real-time updates milte hain
- Lightweight hai (WebSocket se simple)

Use cases:

- Live notifications
- Stock updates
- Chat (simple one-way)
- AI streaming responses (important!)

3. How we use SSE?

- Client connection open karta hai:
- Server stream start karta hai:
- Client continuously sunta rehta hai:

4. Why SSE is Important in AI?

- AI apps (jaise ChatGPT type systems) mein:
- AI apps (jaise ChatGPT type systems) mein:
- Response ek dum nahi aata
  Token by token stream hota hai

SSE ka role:

- Fast streaming responses
- Real-time AI output
- Low complexity vs WebSockets

Server-Sent Events (SSE) Biryani Style
SSE Kya Hai?
Server Khud Update Bhejta Hai Client Ko!
SERVER (Chef)
CLIENT (Customer)
Yahan se updates aayenge!
AI Biryani
Continuous Stream
Push Data: Chawal Ready!
Masala Add Ho Gaya!
Chicken Daal Diya!
Biryani Ready!
Main Aram Se Sun Raha Hoon
No Need to Req bar bar!
Bar Bar Mange: "Biryani Ready Hui?"
Ye Galat Hai!
Request Bharpaar!
SSE Benefits:
Real-Time Updates
Lightweight Connection
Ideal for AI Streaming!
AI Example:
Chef: "Chawal ready ho gaye..."
"Masala add ho gaga..."
"Chicken daal diya..."
"Biryani tayyar!"
ChatGPT Typing Effect
Chef Khud Updates Bhejta Hai... Customer Bas Sunta Rehta Hai!